Worksheet 1: Orbits and Formation

Full solutions

Covers Lectures 1 and 2: history of the field, planet formation, orbital dynamics

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that …"); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end.

Constants and data

$G = 6.674\times10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ $\quad$ 1 AU = 1.496×10^{11} m $\quad$ 1 yr = 3.156×10^{7} s $\quad$ 1 d = 86 400 s

$M_\odot = 1.989\times10^{30}$ kg $\quad$ $M_\oplus = 5.972\times10^{24}$ kg $\quad$ $R_\oplus = 6371$ km

Body	a	P	Mass ($M_\oplus$)	ρ (kg m^{-3})
Mercury	0.387 AU	0.241 yr	0.055	5427
Venus	0.723 AU	0.615 yr	0.815	5243
Earth	1.000 AU	1.000 yr	1.000	5514
Mars	1.524 AU	1.881 yr	0.107	3934
Jupiter	5.203 AU	11.86 yr	317.8	1326
Saturn	9.537 AU	29.46 yr	95.16	687
Uranus	19.19 AU	84.01 yr	14.54	1271
Neptune	30.07 AU	164.8 yr	17.15	1638
Moon	3.844×10^{5} km	27.32 d	0.0123	3344
Io		1.769 d		
Europa		3.551 d		
Ganymede	1.0704×10^{6} km	7.155 d		

Saturn's mean radius is 58 232 km. Planetary and satellite values from NASA/JPL Solar System Dynamics; Galilean satellite periods as quoted in the Lecture 2 notes.

Problem 1 Weighing the Sun and Jupiter

Newton's form of Kepler's third law turns a period and an orbital size into a mass. Here you derive the equation, test a claim against it, and then weigh the Sun with it.

(a) A planet of mass m moves on a circular orbit of radius r about a star of mass M, with $m \ll M$. Starting from the balance between the gravitational force and the centripetal force, derive

$$M = \frac{4\pi^2 r^3}{GP^2}.$$

Solution. The gravitational force supplies the centripetal force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}.$$

The orbital speed on a circular orbit is $v = 2\pi r/P$, so

$$\frac{GMm}{r^2} = \frac{m}{r}\left(\frac{2\pi r}{P}\right)^2 = \frac{m}{r}\frac{4\pi^2 r^2}{P^2} = \frac{4\pi^2 mr}{P^2}.$$

Divide both sides by m:

$$\frac{GM}{r^2} = \frac{4\pi^2 r}{P^2}.$$

Multiply both sides by r^2/G:

$$M = \frac{4\pi^2 r^3}{GP^2}.$$

$M = 4\pi^2 r^3/(GP^2)$**: the orbit radius and period of a satellite measure the mass of the body it orbits.**

(b) Judge this claim in 2 to 3 sentences: *"If Jupiter were twice as massive, its orbital period at its present distance would be noticeably longer."*

Solution. False, on two counts. The planet's mass cancels in part (a): it sets both the gravitational pull and the inertia, so the period contains only the stellar mass and the orbital size, and doubling Jupiter changes nothing at this order. In the exact two-body form, $P^2 = 4\pi^2 a^3/[G(M+m)]$, a heavier planet raises the total attracting mass, so the period becomes slightly *shorter*, not longer; for a second Jupiter mass the change is $\frac{1}{2} M_J/M_\odot \approx 5 \times 10^{-4}$, far below "noticeable".

False: to leading order the planet's mass cancels, and the exact correction is tiny and has the opposite sign.

(c) Compute the mass of the Sun from Jupiter's orbit ($a = 5.203$ AU, $P = 11.86$ yr; show first that $r = 7.784 \times 10^{11}$ m and $P = 3.743 \times 10^8$ s).

Solution. Convert to SI units:

$$r = 5.203 \times 1.496 \times 10^{11}\ \text{m} = 7.7837 \times 10^{11}\ \text{m}, \qquad P = 11.86 \times 3.156 \times 10^7\ \text{s} = 3.7430 \times 10^8\ \text{s}.$$

Form the two powers:

$$r^3 = (7.7837 \times 10^{11})^3 = 4.7158 \times 10^{35}\ \text{m}^3, \qquad P^2 = (3.7430 \times 10^8)^2 = 1.4010 \times 10^{17}\ \text{s}^2.$$

Numerator and denominator separately:

$$4\pi^2 r^3 = 39.478 \times 4.7158 \times 10^{35} = 1.8617 \times 10^{37}\ \text{m}^3,$$

$$GP^2 = 6.674 \times 10^{-11} \times 1.4010 \times 10^{17} = 9.3503 \times 10^6\ \text{m}^3\ \text{kg}^{-1}.$$

Divide:

$$M = \frac{1.8617 \times 10^{37}}{9.3503 \times 10^6} = 1.991 \times 10^{30}\ \text{kg}.$$

The small excess is not an arithmetic slip. The exact two-body form of part (b) returns the *sum* $M_\odot + M_J$, and $M_J/M_\odot \approx 10^{-3}$: the computation weighed the Sun and Jupiter together.

$M_\odot \approx 1.991 \times 10^{30}$ kg, 0.10% **above the accepted** 1.989×10^{30} kg.

Problem 2 The cheapest route to Mars

Treat the orbits of Earth and Mars as circular and coplanar, with radii 1.000 AU and 1.524 AU. A spacecraft transfers between them on an ellipse whose perihelion touches Earth's orbit and whose aphelion touches Mars's orbit. At each end the spacecraft fires its engine briefly (a *burn*), changing its speed by an increment Δv; a burn along the direction of motion is *prograde*. Work entirely in the Sun's gravitational field and ignore the gravity of Earth and Mars themselves.

(a) Two spacecraft of equal mass are on orbits with the same semi-major axis a but different eccentricities. Judge each claim, with justification: (i) *"they have the same total energy"*; (ii) *"they pass through the distance $r = a$ at the same speed"*.

Solution. Both claims are true. (i) The total energy of a Keplerian orbit is $E = -GMm/2a$, which depends on a alone; eccentricity only changes how the fixed total is divided between kinetic and potential energy along the orbit. (ii) Put $r = a$ into the vis-viva equation:

$$v^2 = GM\left(\frac{2}{a} - \frac{1}{a}\right) = \frac{GM}{a}, \qquad \text{so} \qquad v = \sqrt{\frac{GM}{a}},$$

independent of e. On the circular orbit this speed holds everywhere; on the eccentric orbit it holds at the two points where the body is at distance a from the focus, the ends of the minor axis.

Both true; both follow from E depending only on a.

(b) Compute the semi-major axis a_t of the transfer ellipse. Explain why the flight time is half the period of that ellipse, and compute it in days.

Solution. The transfer ellipse has perihelion $r_p = 1.000$ AU and aphelion $r_a = 1.524$ AU. Since $r_p = a(1 - e)$ and $r_a = a(1 + e)$, adding the two gives

$$r_p + r_a = a(1 - e) + a(1 + e) = 2a,$$

so

$$a_t = \frac{r_p + r_a}{2} = \frac{1.000 + 1.524}{2} = 1.262 \text{ AU} = 1.88795 \times 10^{11} \text{ m}.$$

Departure is at perihelion and arrival is at aphelion. These are the two ends of the major axis, so they are separated by half a revolution and the flight takes half a period. With $GM = 6.674 \times 10^{-11} \times 1.989 \times 10^{30} = 1.32746 \times 10^{20}\ \text{m}^3\,\text{s}^{-2}$:

$$a_t^3 = (1.88795 \times 10^{11})^3 = 6.7293 \times 10^{33} \text{ m}^3, \qquad \frac{a_t^3}{GM} = \frac{6.7293 \times 10^{33}}{1.32746 \times 10^{20}} = 5.0693 \times 10^{13} \text{ s}^2,$$

$$P_t = 2\pi\sqrt{5.0693 \times 10^{13}} = 2\pi \times 7.1199 \times 10^6 = 4.4736 \times 10^7 \text{ s}.$$

Half of that is

$$t = \tfrac{1}{2}P_t = 2.2368 \times 10^7 \text{ s} = \frac{2.2368 \times 10^7}{86\,400} = 258.9 \text{ d}.$$

$a_t = 1.262$ AU; **flight time** 259 d, **about** 8.5 **months.**

A faster route to the same number: dividing Kepler's third law for the transfer ellipse by the same law for Earth's orbit (a = 1 AU, P = 1 yr) cancels every constant and leaves $P^2 = a^3$ in units of AU and years, so $P_t = 1.262^{3/2} = 1.418$ yr and $t = 0.709$ yr.

(c) Use the vis-viva equation to show that the spacecraft moves at $v_p = 32.7\ \mathrm{km\,s^{-1}}$ at the transfer perihelion and $v_a = 21.5\ \mathrm{km\,s^{-1}}$ at the transfer aphelion. The circular orbital speeds, from the $r = a$ case of part (a), are $v_\oplus = 29.79\ \mathrm{km\,s^{-1}}$ and $v_{\mathrm{Mars}} = 24.13\ \mathrm{km\,s^{-1}}$. From the four speeds, form the two velocity increments Δv_1 (departure), Δv_2 (arrival), and their sum, and explain why *both* burns point in the direction of motion.

Solution. At perihelion, $r = 1.000$ AU $= 1.496 \times 10^{11}$ m and $a = a_t = 1.88795 \times 10^{11}$ m:

$$\frac{2}{r} = 1.3369 \times 10^{-11}\ \mathrm{m^{-1}}, \qquad \frac{1}{a_t} = 5.2968 \times 10^{-12}\ \mathrm{m^{-1}}, \qquad \frac{2}{r} - \frac{1}{a_t} = 8.0722 \times 10^{-12}\ \mathrm{m^{-1}},$$

$$v_p^2 = GM\left(\frac{2}{r} - \frac{1}{a_t}\right) = 1.32746 \times 10^{20} \times 8.0722 \times 10^{-12} = 1.0716 \times 10^{9}\ \mathrm{m^2\,s^{-2}},$$

$$v_p = 3.2735 \times 10^{4}\ \mathrm{m\,s^{-1}} = 32.73\ \mathrm{km\,s^{-1}}.$$

At aphelion, $r = 1.524$ AU $= 2.2799 \times 10^{11}$ m:

$$\frac{2}{r} = 8.7723 \times 10^{-12}\ \mathrm{m^{-1}}, \qquad \frac{2}{r} - \frac{1}{a_t} = 3.4755 \times 10^{-12}\ \mathrm{m^{-1}},$$

$$v_a^2 = 1.32746 \times 10^{20} \times 3.4755 \times 10^{-12} = 4.6136 \times 10^{8}\ \mathrm{m^2\,s^{-2}},$$

$$v_a = 2.1479 \times 10^{4}\ \mathrm{m\,s^{-1}} = 21.48\ \mathrm{km\,s^{-1}}.$$

The increments:

$$\Delta v_1 = v_p - v_\oplus = 32.73 - 29.79 = 2.94\ \mathrm{km\,s^{-1}},$$

$$\Delta v_2 = v_{\mathrm{Mars}} - v_a = 24.13 - 21.48 = 2.65\ \mathrm{km\,s^{-1}},$$

$$\Delta v_{\mathrm{total}} = 2.94 + 2.65 = 5.59\ \mathrm{km\,s^{-1}}.$$

At departure the transfer demands $32.7\ \mathrm{km\,s^{-1}}$ while Earth's orbit provides 29.8: the spacecraft must *gain* speed, so the first burn is prograde. At arrival the spacecraft crawls at its aphelion speed of $21.5\ \mathrm{km\,s^{-1}}$ while Mars passes at 24.1: the spacecraft is the slower body and must again *gain* speed to match, so the second burn is prograde too.

$\Delta v_1 = 2.94$, $\Delta v_2 = 2.65$, **total** $\Delta v = 5.59\ \mathrm{km\,s^{-1}}$**; both burns add speed along the direction of motion, so both are prograde.**

Problem 3 The Laplace resonance of the Galilean moons

Lecture 2 describes the orbital periods of Io, Europa and Ganymede as standing in a 1 : 2 : 4 ratio. This problem asks how exact that statement is, and what the resonance geometry enforces.

(a) Compute the two consecutive period ratios and the Ganymede-to-Io ratio, and give the percentage by which each misses its integer value.

Solution.

$$\frac{P_{\mathrm{Eur}}}{P_{\mathrm{Io}}} = \frac{3.551}{1.769} = 2.00735, \qquad \frac{2.00735 - 2}{2} = +0.37\%,$$

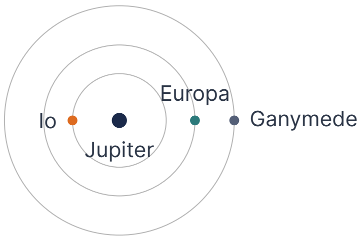
$$\frac{P_{\text{Gan}}}{P_{\text{Eur}}} = \frac{7.155}{3.551} = 2.01493, \qquad \frac{2.01493 - 2}{2} = +0.75\%,$$

$$\frac{P_{\text{Gan}}}{P_{\text{Io}}} = \frac{7.155}{1.769} = 4.04466, \qquad \frac{4.04466 - 4}{4} = +1.12\%.$$

The ratios exceed 2, 2 **and** 4 **by** 0.37%, 0.75% **and** 1.12%.

These offsets are real, not measurement error: the periods are known to far better than a part in 10^4. What the resonance holds exactly is not any pairwise ratio but the *Laplace relation*, a combination of all three mean motions, $n_{\text{Io}} - 3\,n_{\text{Eur}} + 2\,n_{\text{Gan}} \approx 0$, which with these periods vanishes to three parts in 10^5 of n_{Io}, a hundred times more exactly than any ratio above.

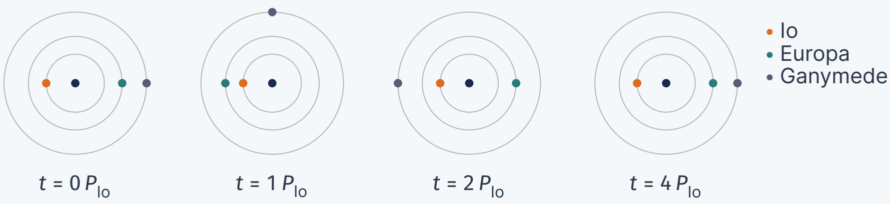
(b) The figure shows the three orbits at the moment of a Europa–Ganymede conjunction; the resonance locks the phases such that Io then stands on the *opposite* side of Jupiter.



Idealise the periods as exactly 1 : 2 : 4. Sketch the positions of the three moons after 1, 2, and 4 Io periods. From your sketch: (i) where does Io stand at every Europa–Ganymede conjunction, and (ii) can a triple conjunction occur?

Solution. Per Io period, Io advances 360°, Europa 180°, and Ganymede 90°. Starting from the printed configuration (Io at 180°, Europa and Ganymede at 0°):

Time (Io periods)	Io	Europa	Ganymede
0	180°	0°	0°
1	180°	180°	90°
2	180°	0°	180°
4	180°	0°	0°



Because Io always advances by full turns per Io period, it returns to 180° at every integer time; Europa alternates between 0° and 180°; Ganymede steps through 90° per Io period. The pattern repeats after 4 Io periods, one Ganymede period.
(i) Europa–Ganymede conjunctions occur at $t = 0, 4, 8, \ldots$, always at 0°, and Io then stands at 180°: at every Europa–Ganymede conjunction, Io is on the opposite side of Jupiter. (ii) Each pairwise conjunction finds the third moon elsewhere: Io–Europa conjunctions (odd t, at 180°) find Ganymede at 90° or 270°; the Io–Ganymede conjunction ($t = 2$, at 180°) finds Europa at 0°.

A triple conjunction never occurs. The resonance phase-protects the system: the moons never line up all on one side, which caps how coherently their mutual perturbations can add.

(c) The resonance holds Io's eccentricity at $e \approx 0.004$ rather than letting tides circularise the orbit. Explain why a non-zero eccentricity is needed for tidal heating to continue in a moon that is already tidally locked, and name the observable consequence.

Solution. A tidally locked moon on a perfectly circular orbit keeps the same face toward the planet at a constant distance. Its tidal bulge is then fixed in the body frame: it neither rises and falls nor moves across the surface, so no work is done deforming the interior and no heat is released.
An eccentric orbit breaks this in two ways at once. The distance to the planet varies over each orbit, so the height of the bulge varies with it. And the moon spins at a constant rate while its orbital angular velocity is fastest at periapsis and slowest at apoapsis, so the direction to the planet librates (oscillates about a mean direction) back and forth in longitude and drags the bulge with it. Both effects flex the interior once per orbit, and internal friction converts that mechanical work into heat.
Left alone, tidal dissipation would damp the eccentricity to zero and switch itself off. The Laplace resonance prevents that by continually re-forcing e.

Observable consequence: Io is the most volcanically active body in the solar system.

Problem 4 Tides and the Roche limit

(a) A moon of radius R_s orbits at distance d from a planet of mass M_p, with $R_s \ll d$. Show that the difference between the planet's gravitational acceleration at the moon's near surface and at its centre is

$$\Delta a \approx \frac{2GM_pR_s}{d^3}.$$

Use $(1 - x)^{-2} \approx 1 + 2x$ for $x \ll 1$.

Solution. The near surface sits at distance $d - R_s$ from the planet, the centre at d:

$$a_{\text{near}} = \frac{GM_p}{(d - R_s)^2}, \qquad a_{\text{centre}} = \frac{GM_p}{d^2}.$$

Factor d out of the first denominator:

$$a_{\text{near}} = \frac{GM_p}{d^2}\left(1 - \frac{R_s}{d}\right)^{-2} \approx \frac{GM_p}{d^2}\left(1 + \frac{2R_s}{d}\right).$$

Subtract:

$$\Delta a = a_{\text{near}} - a_{\text{centre}} = \frac{GM_p}{d^2}\left(1 + \frac{2R_s}{d}\right) - \frac{GM_p}{d^2} = \frac{GM_p}{d^2} \cdot \frac{2R_s}{d} = \frac{2GM_pR_s}{d^3}.$$

The direct pull falls as d^{-2} but the tidal difference falls as d^{-3}, so their ratio is $\Delta a/a = 2R_s/d$: tides are a small correction far away and dominate close in.

$\Delta a = 2GM_pR_s/d^3$ and $\Delta a/a = 2R_s/d$: the tidal difference falls as d^{-3}, one power faster than the direct pull.

(b) The fluid Roche limit is

$$d_R \approx 2.46\, R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3}.$$

Interrogate this formula before using it: (i) why does the *satellite's* radius R_s not appear? (ii) what does it predict as $\rho_s \to \infty$, and does that make sense? (iii) what does it predict for $\rho_s = \rho_p$, and what does that say about how close *any* self-gravitating satellite can orbit?

Solution. (i) Both sides of the comparison scale the same way with satellite size: the tidal acceleration across the satellite, $2GM_pR_s/d^3$, grows linearly with R_s, and so does the satellite's own surface gravity, $Gm_s/R_s^2 = \frac{4}{3}\pi G\rho_s R_s$. The radius cancels in the ratio and only the densities survive. Big and small satellites of the same density are torn apart at the same distance.
(ii) As $\rho_s \to \infty$, $d_R \to 0$: an arbitrarily dense satellite can orbit arbitrarily close, because its self-gravity grows without bound while the tidal stress at fixed d does not. The trend is physically sensible.
(iii) For $\rho_s = \rho_p$ the density factor is unity and $d_R = 2.46\, R_p$: even a satellite as dense as its planet is destroyed within 2.46 planetary radii. The limit falls inside the planet, $d_R < R_p$, only for $\rho_s > 2.46^3\, \rho_p \approx 15\, \rho_p$, far denser than any satellite material, so for every real satellite density the forbidden zone extends well beyond the planet's surface.

R_s cancels because tidal stress and self-gravity both scale linearly with it; only the density ratio matters, and even equal density leaves a forbidden zone of $2.46\, R_p$.

(c) Evaluate d_R for Saturn ($R_p = 58\,232$ km, $\rho_p = 687$ kg m^{-3}) and a satellite of solid water ice ($\rho_s = 1000$ kg m^{-3}). Saturn's main rings span 67 000 to 137 000 km from the planet's centre, with the F ring near 140 000 km: compare. Ring particles are in fact porous aggregates with $\rho_s \approx 500$ kg m^{-3}; *without* recomputing from scratch, state the factor by which the limit moves and what that does to your comparison.

Solution. Solid ice:

$$d_R = 2.46 \times 58\,232 \times 0.8824 = 2.46 \times 51\,384 = 1.264 \times 10^5 \text{ km} = 2.17\, R_p.$$

This sits at 126 400 km: the C, B and inner A rings lie inside it, but the outer A ring (126 400 to 137 000 km) and the F ring lie outside, which the solid-ice limit fails to explain.
Halving the density multiplies d_R by $(1000/500)^{1/3} = 2^{1/3} = 1.26$, because $d_R \propto \rho_s^{-1/3}$:

$$d_R = 1.26 \times 126\,400 = 1.59 \times 10^5 \text{ km} = 2.73\, R_p,$$

and the *entire* ring system, F ring included, now lies inside the limit.

$d_R = 126\,400$ km ($2.17\, R_p$) for solid ice; porosity moves it out by $2^{1/3} = 1.26$ to 159 000 km ($2.73\, R_p$), enclosing the whole ring system.

Problem 5 From planetesimals to planets

(a) Gravitational focusing raises the collision cross-section of a body of radius R to

$$\sigma_{\rm eff} = \pi R^2\left(1 + \frac{v_{\rm esc}^2}{v_\infty^2}\right).$$

Two planetesimals of density 3000 kg m^{-3} have radii 50 km and 500 km. Compute the escape speed of the smaller body, and explain why $v_{\rm esc} \propto R$ at fixed density. Then compute both focusing factors for a swarm with random velocity v_∞ = 10 m s^{-1}, and compare the ratio of the two effective cross-sections with the ratio of the geometric ones.

Solution. For the 50 km body, $R = 5.0 \times 10^4$ m:

$$M = \tfrac{4}{3}\pi R^3\rho = \tfrac{4}{3}\pi(5.0\times10^4)^3 \times 3000 = 1.571\times10^{18}\ \text{kg},$$

$$v_{\rm esc} = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2\times6.674\times10^{-11}\times1.571\times10^{18}}{5.0\times10^4}} = \sqrt{4194} = 64.8\ \text{m s}^{-1}.$$

At fixed density $M \propto R^3$, so $v_{\rm esc} = \sqrt{2GM/R} \propto \sqrt{R^3/R} = R$: the 500 km body has ten times the escape speed, $v_{\rm esc}$ = 648 m s^{-1}.
Focusing factors at v_∞ = 10 m s^{-1}, using $v_{\rm esc}^2$ = 4194 m^2 s^{-2} from above and, since $v_{\rm esc} \propto R$, 100 times that value for the larger body:

$$1 + \frac{4194}{10^2} = 1 + 41.9 = 42.9 \qquad \text{and} \qquad 1 + \frac{4.194\times10^5}{10^2} = 1 + 4194 = 4195.$$

The cross-sections are in the ratio

$$\frac{\sigma_{\rm eff}(500)}{\sigma_{\rm eff}(50)} = \left(\frac{500}{50}\right)^2 \times \frac{4195}{42.9} = 100\times97.8 = 9.8\times10^3,$$

against a geometric ratio of only $(500/50)^2 = 100$.

Focusing gives the larger body an extra factor of 98 on top of its geometric advantage.

(b) Read the limits of the focusing formula: what does $\sigma_{\rm eff}$ reduce to for $v_\infty \gg v_{\rm esc}$, and how does it scale with R for $v_\infty \ll v_{\rm esc}$? Confirm the first limit by evaluating both focusing factors at v_∞ = 500 m s^{-1}. The growing bodies themselves stir the swarm toward their own escape speed: name the two growth regimes these limits correspond to, and explain why the first regime shuts itself down.

Solution. For $v_\infty \gg v_{\rm esc}$ the bracket tends to 1 and $\sigma_{\rm eff} \to \pi R^2$: the purely geometric cross-section, growing only as R^2. For $v_\infty \ll v_{\rm esc}$ the second term dominates,

$$\sigma_{\rm eff} \approx \pi R^2\frac{v_{\rm esc}^2}{v_\infty^2} \propto R^2 \times R^2 = R^4,$$

using $v_{\rm esc} \propto R$ from part (a): the biggest body gains a steeply super-geometric advantage.
At v_∞ = 500 m s^{-1}, with the same $v_{\rm esc}^2$ values:

$$1 + \frac{4194}{500^2} = 1.02, \qquad 1 + \frac{4.194\times10^5}{500^2} = 2.68,$$

so the large body's advantage over pure R^2 scaling collapses from the factor 98 of part (a) to 2.6.

The cold-swarm limit is **runaway growth**: the largest body eats fastest, pulls away from the swarm, and the mass spread widens. The hot-swarm limit is **oligarchic growth**: a few similar-sized bodies grow slowly and in step. The transition is self-inflicted: the winners of runaway growth gravitationally stir the planetesimals, raising v_∞ toward their own v_{esc}, which switches their focusing advantage off.

Geometric πR^2 for a hot swarm, $\propto R^4$ for a cold one; runaway growth creates the stirring that ends it, handing over to oligarchic growth.